



Managed by IHRD, a Govt. of Kerala undertaking

COLLEGE OF ENGINEERING
POONJAR

GROUP 1

PROJECT

SYNAPSE

PREPARED BY

Ihsana Ilyas

Albert Jossy

Abhishek TP

Adinan Azeez KT



REQUIREMENT ANALYSIS & LITERATURE SURVEY

CONTENTS

• Requirement Analysis.....	2
• Functional requirements.....	4
• Non Functional requirements.....	5
• Software requirements.....	5
• Literature survey.....	8
• Competative Summary.....	11
• References.....	13

REQUIREMENT ANALYSIS

Basis in Literature

The Requirement analysis focuses on identifying and documenting all the requirements needed to develop a reliable machine learning solution for predicting train delays in advance. It defines the features the system must provide, the expected level of performance, the software and hardware resources required, and the constraints that influence the project's design and implementation. The requirements are derived from the project objectives, user needs, and insights obtained from existing research on train delay prediction, particularly studies related to Indian Railways. By clearly specifying functional requirements, non-functional requirements, user requirements, system requirements, constraints, and assumptions, this session establishes a structured foundation for the subsequent stages of the project, including the literature survey, feasibility analysis, system design, model development, and implementation. A well-defined requirement analysis ensures that the proposed system addresses the identified problem effectively while remaining practical, scalable, and aligned with the overall project objectives.

Studies consulted while drafting these requirements, together with what each one contributed to a specific requirement below

1. Improving On-Time Performance: Predicting Train Delays with Machine Learning Techniques

Sharma, Hossain, and Kumar (2024) presented a machine learning approach for predicting train delays in the Indian Railways network using the Random Forest classifier. The study demonstrated that Random Forest provides reliable prediction performance when trained on historical train delay data, making it a suitable baseline model for delay prediction tasks. The findings support the inclusion of historical operational data processing (FR-2) and the development of a delay prediction module (FR-4) in the proposed system.

2. Prediction of Train Delay in Indian Railways through Machine Learning Techniques

This study, published through ResearchGate/IJRTE, evaluated several machine learning algorithms, including K-Nearest Neighbours (KNN), Decision Tree, Logistic Regression, and Random Forest, for predicting train delays in Indian Railways. The research also identified festival-driven overcrowding as one of the significant factors influencing train delays. These findings justify incorporating weekends, public holidays, and religious festivals as contextual input features in the proposed system, supporting FR-3 (Contextual Data Integration).

3. Train Delay Estimation in Indian Railways by Including Weather Factors Through Machine Learning Techniques

This research focused on improving train delay prediction by integrating weather-related information into the prediction model. The authors employed ensemble learning methods such as AdaBoost, Gradient Boosting, and XGBoost to analyse the impact of weather conditions on train punctuality. The study demonstrates the importance of weather as a predictive feature and supports the inclusion of automatic weather data integration (FR-3) as well as the evaluation of boosting-based machine learning algorithms in the proposed system.

4. Modeling Train Delays: A Study of Indian Railways

Published in ScienceDirect (2025), this study compared both regression and classification approaches for train delay prediction using Indian Railways data. The results showed that Extra Trees Regression and an XGBoost classifier achieved superior prediction performance compared to other models. Rather than recommending a single algorithm, the study highlights the importance of evaluating multiple tree-based ensemble methods before selecting the most suitable model. This directly supports the model selection strategy adopted in the proposed project.

4. A Novel Deep Learning Model for Short-Term Train Delay Prediction

This ScienceDirect (2023) study proposed a WT-FCNN-LSTM deep learning model for short-term train delay prediction and compared its performance against Random Forest, Gradient Boosted Regression Trees (GBRT), XGBoost, and Support Vector Regression (SVR) using Chinese high-speed railway data. The results showed that while deep learning models achieved excellent performance, tree-based ensemble methods remained highly competitive. This finding supports the decision to begin the proposed project with Random Forest, XGBoost, and LightGBM before considering more computationally intensive deep learning approaches.

6. Real-Time Passenger Train Delay Prediction Using Machine Learning: An Amtrak Case Study

This IEEE (2022) study investigated train delay prediction using historical operational records together with real-time contextual information. The authors compared Random Forest, Gradient Boosting, and Multi-Layer Perceptron (MLP) models and demonstrated that combining historical and real-time data significantly improves prediction accuracy. These findings support the integration of historical train records (FR-2) and contextual information such as weather and calendar data (FR-3) in the proposed Train Delay Prediction System.

8. Predicting Train Delays Using a Two-Level Machine Learning Approach

Presented at ICAART (2022), this research proposed a two-stage LightGBM framework consisting of a delay classification model followed by a regression model that predicts the exact delay duration. The study introduced the concept of estimating the likelihood of delay before calculating precise delay minutes, thereby improving prediction confidence. This approach inspired the inclusion of FR-5 (Confidence Score Generation) in the proposed system.

8. RSTGCN: Railway-Centric Spatio-Temporal Graph Convolutional Network for Train Delay Prediction

The arXiv (2025) study proposed a Railway-Centric Spatio-Temporal Graph Convolutional Network (RSTGCN) for station-level train delay prediction across the Indian Railways network. The research reported an average arrival delay of approximately 51 minutes during September 2024, highlighting the widespread nature of train delays in India. These findings establish the practical importance of developing an accurate prediction system and provide motivation for the project's accuracy objective described in NFR-1.

9. Flight Delay Prediction Using Machine Learning and Explainable AI

Although this ScienceDirect (2025) study focuses on flight delays at Hazrat Shahjalal International Airport, it demonstrates the application of Explainable Artificial Intelligence (XAI) using SHAP values to interpret machine learning predictions. The research emphasizes the importance of explaining why a prediction is made rather than only presenting the prediction itself. This supports FR-6 (Explanation of Prediction) by motivating the use of feature importance and explainability techniques in the proposed Train Delay Prediction System.

10. Passenger Demand Forecasting for Railway Systems

This 2022 study investigated passenger demand forecasting using artificial neural networks and other machine learning techniques for railway systems. Although the primary focus is passenger demand rather than train delays, the research demonstrates that machine learning can effectively support operational planning in railway transportation. This provides a strong foundation for the proposed future enhancement (FR-8), where passenger demand prediction can be integrated alongside train delay prediction to improve railway planning and resource allocation, quickly and easily.

FUNCTIONAL REQUIREMENTS

1. Journey Input

The system shall accept the train number, journey date, boarding station, and destination station from the user.

2. Historical Data Processing

The system shall process historical running data, including scheduled and actual arrival/departure times and recorded delay information, for the selected trains.

3. Contextual Data Integration

The system shall automatically retrieve and integrate weather forecast information and calendar data, including weekends, public holidays, and religious festivals, for the selected journey date without requiring manual input from the user.

4. Delay Prediction

The system shall predict the expected delay, measured in minutes, at each station along the selected train route.

5. Confidence Score Generation

The system shall generate a confidence score for every predicted delay to indicate the reliability of the prediction.

6. Explanation of Prediction

The system shall identify and display the key factors contributing to each predicted delay so that users can understand the basis of the prediction.

7. Results Display

The system shall present the predicted delay, confidence score, and contributing factors in a clear, readable, and user-friendly format.

8. Passenger Demand Prediction (Future)

The system shall be extensible to forecast passenger demand alongside train delay prediction to support railway operational planning.

9. Model Retraining

The system shall support periodic retraining of the prediction model as new operational data becomes available to improve prediction accuracy over time.

NON FUNCTIONAL REQUIREMENTS

Non-functional requirements describe how well the system must perform its functions, rather than the functions themselves.

1. Accuracy

Predictions shall be evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the R^2 score. Given that Indian Railways data has been reported to show average arrival delays in the tens of minutes across the network, the model should meaningfully reduce uncertainty for a specific train and station rather than target exact-minute precision.

2. Reliability

The system shall return consistent predictions for identical inputs and degrade gracefully, rather than failing outright, when contextual data such as weather information is temporarily unavailable.

3. Scalability

The underlying machine learning pipeline shall be designed so that it can be extended from the initial two to three pilot trains to a larger number of trains without requiring architectural changes.

4. Security

User input and any stored records shall be handled securely without exposing raw datasets or personal information.

5. Usability

The user interface shall be simple enough for passengers with no technical background and shall require only journey details as input.

6. Performance

The system shall generate and return a prediction within a few seconds under normal operating conditions.

7. Availability

The web application shall remain accessible with minimal downtime during normal operation.

SOFTWARE REQUIREMENTS

1 .Software Requirements

Python is specified as the core language rather than an alternative such as R or Julia. This is a deliberate choice, not a default: Table 2.4a summarises the trade-off. R remains strong for pure statistical analysis and reporting, and Julia offers better raw numerical performance, but neither matches Python's combination of ML libraries (Scikit-learn, XGBoost, LightGBM), deployment tooling (Flask/FastAPI), and community support — and every ML-for-train-delay paper reviewed in Section 2.1 that names its tooling builds on this same Python ecosystem, which is itself a practical signal for a project in this exact domain.

Software language comparison

Language	Status	Rationale
Python	Selected	Largest ML ecosystem (Scikit-learn, XGBoost, LightGBM, Pandas). native Flask/FastAPI deployment.largest community support for troubleshooting.
R	Not selected	Excellent for statistical analysis, visualization (ggplot2), and reporting; remains the norm in biostatistics and academic statistics, but has a smaller ML-deployment ecosystem and less first-party support for XGBoost/LightGBM workflows and web deployment.
Julia	Not selected	Strong raw numerical performance, useful for heavy simulation; however its ML and web-deployment ecosystem is far less mature, with fewer tutorials, smaller community, and limited first-party support for the libraries this project depends on.

Software Requirements

Category	Software/Tool
Peogramming Language	Python 3.x
Experimentation Environment	Google Collab / Jupyter Notebook
Devolopment IDE	Visual Studio Code
Data Handling	Pandas,Numpy
Baseline Machine Learning	Scikit-learn
Advanced Regression Models	XGBoost,LightGBM
Backend API	Flask(Streamlit as an optional rapid-prototyping alternative)
Version Control	Git and GitHub

2. Hardware Requirements

Component	Recommended Specialization
Processor	Intel Core i5 or equivalent (i7 or higher recommended for faster model training)
Ram	8 GB minimum; 16 GB recommended when working with larger historical datasets
Storage	At least 5–10 GB of free space for datasets, trained model artifacts, and the development environment
Internet Connection	Required for weather-API calls, holiday/festival data retrieval, and any cloud-based training or deployment

3. User Requirements

- A user should be able to obtain a prediction using only journey details — train number, date, boarding and destination station — without needing any technical knowledge of how the system works.
- A user should receive a response within a short, predictable time frame.
- A user should be able to see why a delay is predicted, not just the number itself, so the prediction can be trusted and acted on.

4. System Constraints

- The initial version is limited to two to three selected trains, based on the volume of historical data that can realistically be collected and cleaned within the project timeline.
- Prediction accuracy is bounded by the quality and completeness of the historical data available for those trains.
- Contextual data — weather forecasts and holiday/festival calendars — depends on the continued availability of third-party APIs and public datasets.
- The system is advisory in nature; it has no control over, and cannot directly influence, actual railway operations.

5. Assumptions

- Historical running data for the selected trains can be obtained in sufficient volume and detail to train a reliable model.
- Weather and holiday/festival data sources used during development remain accessible for the duration of the project.
- Users of the web application have a working internet connection.
- Some prediction error is unavoidable, since real-world disruptions such as accidents or equipment failure cannot be anticipated from historical patterns alone.

LITERATURE SURVEY

Search Strategy

The survey was carried out against the search terms identified in Section 2.1: train delay prediction, machine learning in railways, railway delay analysis, AI in transportation, passenger demand prediction, and explainable AI (XAI). Searches were run across Google Scholar, IEEE Xplore, ScienceDirect, Springer, and arXiv, and preference was given to work published in the last five years. A small number of older papers (2019–2021) are retained where they are Indian-Railways-specific and there is no more recent equivalent covering the same ground for a network as under-studied in the ML literature as Indian Railways, excluding them would leave the survey with almost no domestic baseline to compare against.

Ten papers were selected for detailed review: four addressing Indian Railways directly, three addressing delay prediction on other national networks (China, the United States, Tunisia), one addressing full-network station-level delay prediction at a scale not yet attempted for India, one applying explainable AI to delay prediction in an adjacent transportation domain (aviation), and one addressing railway passenger demand forecasting. This spread was chosen deliberately: the Indian-context papers ground the project in its actual operating environment, the international papers show what more mature research pipelines look like, and the demand-forecasting and explainability papers support two of the project's forward-looking objectives (FR-8 and FR-6 respectively).

Reviewed Literature

Indian Railways -Delay Prediction

1.Sharma, Hossain & Kumar (2024) — Improving On-Time Performance: Predicting Train Delays with Machine Learning Techniques.

- Problem addressed: forecasting train delays on the Indian Railways network to help both passengers and railway authorities identify operational bottlenecks.
- Dataset: real-world Indian Railways operational data.
- Algorithm: Random Forest Classifier.
- Result: proposed as a practical decision-support tool for both journey planning and bottleneck identification; the full performance figures sit behind the publisher paywall and were not accessible for this review.
- Limitation: only one algorithm family (Random Forest) is reported in the available abstract, with no comparison against boosting or deep-learning alternatives.

2.Arshad & Ahmed (2019) — Prediction of Train Delay in Indian Railways through Machine Learning Techniques.

- Problem addressed: identifying the predictors of delay on Indian Railways, explicitly including festival-driven overcrowding as a contributing factor.
- Dataset: real-time data collected from an Indian Railways regional office.
- Algorithm: k-Nearest Neighbours, Decision Tree, Logistic Regression, and Random Forest — used for classification (delayed vs. not delayed) rather than predicting exact minutes.
- Result: identified festival crowding, alongside infrastructure and population factors, as a material contributor to delay; Random Forest was the strongest of the four classifiers tested.
- Limitation: predicts whether a train will be delayed, not the size of the delay, and is based on data from a single regional office rather than a network-wide sample.

3.Arshad & Ahmed (2021) — Train Delay Estimation in Indian Railways by Including Weather Factors Through Machine Learning Techniques.

- Problem addressed: estimating the magnitude of delay (rather than just its presence) once weather is added as a feature.
- Dataset: Indian Railways operational records with associated weather data.
- Algorithm: ensemble stacking combining AdaBoost, Gradient Boosting, and XGBoost regressors.

- Result: identified festival crowding, alongside infrastructure and population factors, as a material contributor to delay; Random Forest was the strongest of the four classifiers tested.
- Limitation: predicts whether a train will be delayed, not the size of the delay, and is based on data from a single regional office rather than a network-wide sample.

4. Arshad & Ahmed (2021) — Train Delay Estimation in Indian Railways by Including Weather Factors Through Machine Learning Techniques.

- Problem addressed: estimating the magnitude of delay (rather than just its presence) once weather is added as a feature.
- Dataset: Indian Railways operational records with associated weather data.
- Algorithm: ensemble stacking combining AdaBoost, Gradient Boosting, and XGBoost regressors.
- Result: demonstrates that stacked ensembles of boosting algorithms are a workable approach once weather is included as a feature, extending the same authors' 2019 classification work into a regression setting.
- Limitation: as a direct extension of the 2019 study, it inherits the same single-region data constraint, and the abstract available for review does not report a specific error metric.

5. Banerjee, Bhattacharyya, Sharma & Panja (2025) — Modeling Train Delays: A Study of Indian Railways.

- Problem addressed: identifying delay patterns over time across multiple Indian Railways zones, comparing regression- and classification-based modelling approaches.
- Dataset: delay data from 20 trains collected across multiple zones between 2015 and 2019, provided by South Eastern Railways.
- Algorithm: multiple regression and classification models compared, including Extra Trees Regression and an XGBoost classifier.
- Result: Extra Trees Regression and the XGBoost classifier outperformed the other models tested.
- Limitation: the authors themselves flag the need for pan-India coverage (their sample is 20 trains across a subset of zones) and for online/real-time algorithms, which this project's initial two-to-three-train pilot scope shares.

International Rail Delay Prediction

1. Gao, Zhang, Ou & Dong (2023) — A Novel Deep Learning Model for Short-Term Train Delay Prediction.

- Problem addressed: short-term delay prediction that reacts to a train's evolving real-time state, rather than a purely historical-pattern forecast.
- Dataset: real-world operational data from two Chinese high-speed rail lines.
- Algorithm: a hybrid Wavelet Transform + Fully Connected Neural Network + LSTM model ("W-L-F"), benchmarked against Random Forest, Gradient Boosting Regression Trees, XGBoost, and Support Vector Regression.
- Result: the hybrid deep-learning model outperformed the tree-based and SVR baselines on short-term prediction accuracy.
- Limitation: built for double-track, high-speed, relatively punctual Chinese rail operations — a materially different operating environment from the mixed-traffic, higher-variance Indian Railways context this project targets.

2. Lapamonpinyo, Derrible & Corman (2022) — Real-Time Passenger Train Delay Prediction Using Machine Learning: A Case Study With Amtrak Passenger Train Routes.

- Problem addressed: real-time delay prediction for US passenger rail, where trains often run on freight-owned and freight-dispatched track.
- Dataset: Amtrak passenger train routes (routes R364, R370, and R350).
- Algorithm: Random Forest, Gradient Boosting Machine, and Multi-Layer Perceptron, compared under two input structures — real-time-only data, and real-time data combined with historical delay profiles.
- Result: the MLP model using the combined real-time-plus-historical data structure outperformed all other model/structure combinations; external variables such as ridership, population, day of week, geography, and weather were shown to materially improve predictions.
- Limitation: US freight-dispatched track operations differ structurally from Indian Railways; the finding that combining historical and real-time features improves accuracy is nonetheless directly transferable and supports FR-2/FR-3.

3. Laifa, Khcherif & Ben Ghezala (2022) — Predicting Trains Delays Using a Two-Level Machine Learning Approach.

- Problem addressed: predicting passenger train delay on the Tunisian railway in a way that is both accurate and easy to interpret as a delay "band."
- Dataset: Tunisian National Railways Company (SNCFT) records — 12,350 travel samples from 1 January to 31 December 2019, covering 55 passenger trains and 4 main destinations.
- Algorithm: a novel two-level LightGBM approach — a first-level classifier assigns the journey to one of 13 five-minute delay bands, and a second-level regressor then predicts the exact number of minutes within that band.
- Result: the two-level approach outperformed both a single-level LightGBM model and other one-level benchmarks (including Random Forest and a neural network).
- Limitation: tested on a comparatively small national network (55 trains, one year of data); the two-level classify-then-regress design is nonetheless a strong candidate pattern for this project's own confidence-scoring requirement (FR-5).

Full-Network, Station-Level Delay Prediction

1. Chowdhury, Koley, Chakraborty & Ghosh (2025) — RSTGCN: Railway-Centric Spatio-Temporal Graph Convolutional Network for Train Delay Prediction.

- Problem addressed: forecasting average hourly arrival delay at every station across the entire Indian Railways network, rather than delay for one train at a time.
- Dataset: the first nationwide Indian Railways operational dataset assembled for this purpose — 3,892 long-distance trains and 4,735 stations across 17 zones, covering September 2024 in full.
- Algorithm: a graph convolutional network with a spatio-temporal attention mechanism and a novel "hourly headway" feature, benchmarked against Historical Average, Random Forest, LSTM, GRU, STGCN, ASTGCN, and TSTGCN.
- Result: RSTGCN outperformed every baseline on MAE, MAPE, and RMSE across nearly all 17 railway zones; separately, the authors' analysis of September 2024 data found a network-wide average arrival delay of roughly 51 minutes, with some delays exceeding 12 to 24 hours.
- Limitation: predicts average delay per station per hour rather than the delay of a specific train a specific passenger has booked, and the architecture is considerably more complex to build and maintain than the tree-based models this project starts with; the 51-minute network average is nonetheless a useful benchmark for what "a meaningful improvement in certainty" should mean in NFR-1.

Explainability in Delay Prediction (Cross-Domain)

1. Chowdhury, Mahmud, Mashruk, Basunia, Hasib & Alam (2025) — Flight Delay Prediction Using Machine Learning and Explainable AI: A Case Study on Hazrat Shahjalal International Airport, Dhaka.

- Problem addressed: most delay-prediction studies report accuracy only and do not explain why a delay is predicted — this paper closes that gap for flight delay in a developing-country airport context.
- Dataset: 2022 flight and weather data for Hazrat Shahjalal International Airport, Dhaka.
- Algorithm: KNN, SVM, Decision Tree, Random Forest, AdaBoost, XGBoost, and CatBoost, paired with SHAP-based explainable AI.
- Result: combining strong predictive models with SHAP explanations let the authors show which weather and operational features drove individual predictions, not just how accurate the model was overall.
- Limitation: aviation rather than rail, and delay dynamics differ between the two modes; the explainability approach — pairing a boosting model with SHAP-style feature attribution — is the direct template for this project's FR-6 (explanation of prediction).

Passenger Demand Forecasting

2. Nar & Arslankaya (2022) — Passenger Demand Forecasting for Railway Systems.

- Problem addressed: forecasting rail passenger demand at two levels of granularity — an overall line-level forecast and a more detailed station-level forecast.
- Dataset: ridership records for a metro/railway line, used to build both line-basis and station-basis forecasts.
- Algorithm: regression analysis and a simple average for the line-level forecast; artificial neural networks and other machine learning algorithms for the station-level forecast.

- Result: the station-level ANN/ML approach achieved a mean absolute percentage error below 2%, outperforming the simpler line-level statistical methods.
- Limitation: demand forecasting and delay forecasting are treated as separate problems here, with no shared feature pipeline between them; this project's FR-8 (future passenger demand prediction) would need to establish that link deliberately rather than assume it.

Comperative Summary

Algorithm and dataset comparison

Title (short)	Year	Algorithm	Dataset
Improving On-Time Performance (Sharma et al.)	2024	Random Forest Classifier	Indian Railways operational data
Prediction of Train Delay in Indian Railways (Arshad & Ahmed)	2019	KNN, Decision Tree, Logistic Regression, Random Forest	Indian Railways regional-office data
Train Delay Estimation... Weather Factors (Arshad & Ahmed)	2021	Stacked ensemble: AdaBoost, Gradient Boosting, XGBoost	Indian Railways data + weather
Modeling Train Delays (Banerjee et al.)	2025	Extra Trees Regression, XGBoost Classifier (best of several compared)	20 Indian trains, multiple zones, 2015–2019
A Novel Deep Learning Model (Gao et al.)	2023	WT + FCNN + LSTM hybrid vs. RF, GBRT, XGBoost, SVR	2 Chinese high-speed rail lines
Real-Time PTDP — Amtrak (Lapamonpinyo et al.)	2022	Random Forest, GBM, MLP (RT-DFS vs. RWH-DFS)	Amtrak routes R364, R370, R350
Two-Level ML Approach (Laifa et al.)	2022	Two-level LightGBM (classify band, then regress)	SNCFT Tunisian Railway, 12,350 samples, 2019
RSTGCN (Chowdhury et al.)	2025	Graph Convolutional Network + spatio-temporal attention	Full Indian Railway network, Sept 2024
Flight Delay Prediction + XAI (Chowdhury et al., Dhaka)	2025	RF, XGBoost, CatBoost, etc. + SHAP explainability	Dhaka airport, 2022 flight + weather data
Passenger Demand Forecasting (Nar & Arslankaya)	2022	Regression (line-level); ANN/ML (station-level)	Metro/railway ridership records

Result and Limitation Comparison

Title (short)	Key Result / Strength	Limitation
Improving On-Time Performance	Practical decision-support framing for both passengers and operators	Single algorithm reported; full results paywalled

Title (short)	Key Result / Strength	Limitation
Prediction of Train Delay (2019)	First to link festival crowding to delay in an Indian context	Classification only (delayed / not), single region
Weather Factors (2021)	Extends classification into minute-level regression	Same single-region constraint as the 2019 study
Modeling Train Delays (2025)	Extra Trees / XGBoost outperform other candidates tested	Authors flag need for pan-India, real-time coverage
Novel Deep Learning Model	Deep model beats tree baselines on short-term accuracy	Built for punctual, double-track high-speed rail
Amtrak PTDP	Shows historical+real-time features beat real-time-only	US freight-dispatched track differs from Indian context
Two-Level ML (Tunisia)	Classify-then-regress beats one-level models	Small network, one year of data
RSTGCN	State-of-the-art on full-network Indian station delay	Predicts station averages, not a specific train's delay
Flight Delay + XAI (Dhaka)	Pairs strong accuracy with SHAP-based explanation	Aviation, not rail; single airport
Passenger Demand Forecasting	Station-level ANN/ML achieves under 2% MAPE	Treats demand and delay as unconnected problems

Research Gap Analysis

Read together, the ten papers point to five consistent gaps in the existing literature:

- Limited contextual features: most Indian-context studies (Arshad & Ahmed, 2019; Banerjee et al., 2025) work with a narrow feature set — typically delay history plus one contextual factor such as weather or festivals — rather than combining weather, weekends, holidays, and festivals together in a single model.
- Weather and festival analysis in the same model: weather (Arshad & Ahmed, 2021) and festival crowding (Arshad & Ahmed, 2019) have so far been studied separately in the Indian context, not jointly, despite both being flagged as material delay factors.
- Limited explainable AI in rail delay prediction specifically: the only paper reviewed that pairs delay prediction with a structured explanation of individual predictions (SHAP-based feature attribution) is in aviation (Chowdhury et al., Dhaka, 2025), not rail — none of the seven rail-delay papers reviewed report an equivalent explainability layer.
- No integrated passenger demand prediction: passenger demand forecasting (Nar & Arslankaya, 2022) and delay prediction are addressed as entirely separate research problems in every paper reviewed; none combine the two in a single pipeline a passenger-facing system could use.
- Limited real-time integration at India's network scale: RSTGCN (Chowdhury et al., 2025) is the only paper operating at full Indian-network scale, and it predicts hourly station averages rather than a specific train's delay on a specific booked journey — the finer-grained, journey-level, real-time prediction this project targets has not been demonstrated at that scale.

REFERENCES

1. R. C. Sharma, I. Hossain, and A. Kumar, "Improving On-Time Performance: Predicting Train Delays with Machine Learning Techniques," in Proc. Int. Conf., Springer, 2024.
2. "Prediction of Train Delay in Indian Railways through Machine Learning Techniques," ResearchGate.
3. "Train Delay Estimation in Indian Railways by Including Weather Factors Through Machine Learning Techniques," ResearchGate.
4. "Modeling Train Delays: A Study of Indian Railways," ScienceDirect, 2025.
5. "A Novel Deep Learning Model for Short-Term Train Delay Prediction," ScienceDirect, 2023.
6. "Real-Time Passenger Train Delay Prediction Using Machine Learning: A Case Study With Amtrak Passenger Train Routes," IEEE Xplore, 2022.
7. "Predicting Trains Delays Using a Two-Level Machine Learning Approach," in Proc. 14th Int. Conf. Agents and Artificial Intelligence (ICAART), 2022.
8. "RSTGCN: Railway-Centric Spatio-Temporal Graph Convolutional Network for Train Delay Prediction," arXiv, 2025.
9. "Flight Delay Prediction Using Machine Learning and Explainable AI: A Case Study on Hazrat Shahjalal International Airport, Dhaka," ScienceDirect, 2025.
10. "Passenger Demand Forecasting for Railway Systems," 2022.